

Central venoarterial extracorporeal life support in pediatric refractory septic shock: a single center experience

Perfusion

2022, Vol. 37(4) 385–393

© The Author(s) 2021

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/02676591211001782

journals.sagepub.com/home/prf



Amanda Ruth,¹  Adam M Vogel,² Iki Adachi,³
Lara S Shekerdeman,¹ Patricia Bastero¹ and James A Thomas¹

Abstract

Objective: Venoarterial extracorporeal membrane oxygenation (VA ECMO) is recognized as a potential support therapy for pediatric patients with refractory septic shock (RSS). This review aims to report our experience with central VA cannulation in pediatric patients with RSS, and to compare this with peripheral VA ECMO cannulations for this condition at our institution.

Design: Retrospective case series.

Setting: Pediatric and cardiac intensive care units in an academic pediatric hospital.

Patients: All patients 0–18 years old meeting criteria of RSS placed on VA ECMO between January 2011 and December 2018.

Interventions: None.

Measurements: Demographics, relevant clinical variables, ECMO run details, and outcomes were collected.

Results: Between 2011 and 2018, 14 children were placed on VA ECMO for RSS. Nine were cannulated centrally, with the rest placed on peripheral VA ECMO. Overall survival to hospital discharge was 57.1% (8/14), with 66.7% of the central cannulation cohort surviving versus 40% in the peripheral cannulation ($p=0.34$). Median ECMO duration was 147.1 hours (IQR: 91.9–178.6 hours), with survivors having a median length of 147.1 (IQR: 138.5–185.7) versus non survivors 114.7 hours (IQR: 63.7–163.5), $p=0.48$. Overall median ICU length of stay (LOS) was 19 days (IQR: 10.5–42.2). The median % maximum flow achieved on VA ECMO was higher in the central cannulation group at 179.6% (IQR: 154.4–188.1) versus the peripheral with 133.5% (98.1–149.1), $p=0.01$. Functional status scale (FSS) was used to capture morbidity. All survivors had a mean increase in their FSS from baseline. In the centrally cannulated group, 50% (4/8) received mediastinal exploration, but none developed mediastinitis. In terms of blood product utilization, the central cannulation received more platelets compared to the peripherally cannulated group (median 15.6 vs 3.3 mL/kg/day, $p=0.03$).

Conclusion: A central approach to VA ECMO cannulation is feasible and has potential for good patient outcomes in selected patients.

Keywords

pediatric refractory septic shock; pediatric extracorporeal membrane oxygenation; central cannulation

Introduction

Septic shock, according to the most recent Surviving Sepsis Campaign definition, refers to a subset of sepsis where severe infection leads to cardiovascular dysfunction.¹ The mortality from septic shock has been reported to be between 15% and 49% depending on the presentation.^{2–4} While the majority of patients respond to conventional medical interventions, such as early source control and standard consensus-driven medical

¹Division of Pediatric Critical Care Medicine, Department of Pediatrics, Texas Children's Hospital and Baylor College of Medicine, Houston, TX, USA

²Division of Pediatric Surgery, Department of Surgery, Texas Children's Hospital and Baylor College of Medicine, Houston, TX, USA

³Division of Congenital Heart Surgery, Department of Surgery, Texas Children's Hospital and Baylor College of Medicine, Houston, TX, USA

Corresponding author:

Amanda Ruth, Division of Pediatric Critical Care Medicine, Department of Pediatrics, Texas Children's Hospital and Baylor College of Medicine, 6651 Main Street, Houston, TX 77030, USA.

Email: axruth@texaschildrens.org

interventions including fluid resuscitation, vasoactive medication, and respiratory support, there exists a minority of patients that remain in decompensated, “refractory” shock with ongoing organ dysfunction. This subset of patients have a mortality rate of up to 50%^{4,5} despite aggressive treatment according to consensus recommendations.^{6,7}

The most recent American College of Critical Care Medicine-Pediatric Advanced Life Support (ACCM-PALS) algorithm for the treatment of pediatric septic shock includes veno-arterial (VA) extracorporeal membrane oxygenation (ECMO) as a potential rescue therapy for patients with refractory shock.⁸ The 2020 update of the Surviving Sepsis Campaign guidelines for septic shock management similarly suggests VA ECMO as a rescue.⁹ Early experience with peripheral (neck or femoral vessel) cannulation for VA ECMO for septic shock revealed disappointing outcomes with reported mortality of up to 83%.^{10,11} More recently, in the largest cohort of pediatric septic shock patients examined to date, Oberender et al. reported an overall survival rate of 50% in children across seven international pediatric intensive care units (PICU)¹² between 2006 and 2014. While this represents a significant improvement over the previously reported mortality rates of 80%, it falls short of ECMO survival results for other conditions, such as ARDS.

This finding suggests that, at a time when the body's oxygen demands are substantially elevated, it is possible that the circulatory demands may not be fully met by the flows that the conventional peripheral ECMO is able to provide. The mismatch between oxygen consumption and delivery leads to ongoing tissue dysoxia with organ dysfunction. Tissue dysoxia is further aggravated by the presence of microcirculatory shunting characteristic to severe septic shock, which oftentimes manifests as disproportionately high mixed venous saturations due to oxygen extraction failure. To address such delivery mismatch and remedy tissue dysoxia, ECMO with central cannulation (draining from the right atrium and returning to the aorta) has been suggested as a potential therapeutic option. High-flow capability of the central approach allows for weaning vasoconstrictors (and even starting vasodilators) while still maintaining an appropriate systemic blood pressure which is critical to vital organs such as the heart and brain. A single Australian institution has reported a 73% survival to hospital discharge in pediatric patients with refractory shock who were cannulated centrally for ECMO.^{13,14} A more recent review of VA ECMO for children with sepsis similarly found central cannulation as a significant predictor for survival.¹⁵ Of note, despite an overall survival of 50%, Oberender et al.¹² noted that within their cohort, patients who had received higher ECMO flows (>150 mL/kg/min) as opposed to standard flows had a

much higher survival rate of 82%. In 2011, our institution chose to adopt a central cannulation approach to ECMO support for refractory septic shock for patients in whom this was feasible and not contraindicated. In this report we present our experience to date with this approach.

Material and methods

A retrospective review of all patients ages 0–18 receiving ECMO in the pediatric and cardiac ICUs (PICU and CICU) at Texas Children's Hospital from January 2011 to December 2018 was performed. The Baylor College of Medicine Institutional Review Board granted permission to conduct this research and waived the requirement for consent given its retrospective nature. Children with a primary diagnosis of severe sepsis/septic shock and refractory vasoplegic shock were included if they were cannulated for central or peripheral VA ECMO. While we do not have strict criteria for central versus peripheral cannulation, relative contraindications to central cannulation during this period were profound underlying immune suppression, resuscitation at the time of cannulation or the presence of multiple prior sternotomies. Cannulation is performed at the bedside for peripheral VA ECMO, and in the cardiac operating room for central VA ECMO. Once on ECMO, medical management is according to institutional protocols and includes rigorous monitoring of coagulation status, bleeding, with blood product replacement according to laboratory parameters and clinical status. Flows are maintained to optimize markers of systemic perfusion, with a low threshold for continuous fluid removal, continuous renal replacement therapy for control of fluid balance, renal support, and management of profound inflammation. Our center also routinely utilized plasmapheresis as adjunctive therapy for septic patients with thrombocytopenia-associated multiple organ failure (TAMOF).

Demographics, comorbidities, and the presence of cardiac arrest pre-ECMO were recorded, as well as additional variables during the ECMO run including: highest arterial lactates, selected laboratory values, vasotropic inotropic scores (VIS) prior to ECMO and at 6 and 24 hours of ECMO, use of vasodilators after ECMO initiation, the use of adjunctive extracorporeal therapies, highest ECMO flows achieved throughout a patient's VA ECMO run. Given the wide variations in the age and size of the patients included, ECMO flow is described as a percentage relative to what is supposed to be “full-flow” for each patient (e.g. 200% if the ECMO flow achieved is double the expected full flow). While our institution does not mandate an absolute number of what is considered “full-flow,” we have guidelines on a starting number based on the age and body-surface area

of the patient¹⁶ (Supplemental Material), with flow subsequently adjusted to meet clinical endpoints. Outcomes of interest included bleeding complications, neurologic status, hospital and ICU length of stay (LOS), and mortality. Neurologic outcome was assessed using functional status scale (FSS)¹⁷ that were collected on admission and at discharge. Functional status scale is a validated assessment of neurologic status¹⁸ and has been increasingly used in pediatric studies.^{19,20} Descriptive statistical analysis was performed. Results are reported as median (IQR) or mean (SD) as appropriate. As the cohort is small and non-parametrically distributed, the Mann-Whitney *U* test was used to compare continuous variables, whereas Chi-square was used for categorical ones.

Given that central cannulation for septic shock is still a rarely used technique for pediatric refractory septic shock in the United States, we also compared our mortality rates with those of other centers in the Extracorporeal Life Support Organization (ELSO) database. The database was queried for any ECMO patient aged 0–18 year old from January 2011 to December 2018 with the diagnosis of septic shock. Patients were selected if they had an ICD-9 code of 995.91, 995.92, 785.52, and/or an ICD-10 code of R65.21 either as primary or secondary diagnoses. The cohort was further refined to patients with septic shock or meningococemia as primary diagnosis, and those with congenital heart disease excluded. Those listed under the mode of “VV, other, unknown” and with support indicated as “pulmonary” were excluded as well. Patients were identified as having been centrally cannulated if cannulae were recorded as having been placed in the aorta and right atrium.

Results

Between January 2011 and December 2018, 14 patients were placed on VA ECMO for refractory septic shock. Nine were cannulated centrally (CC), with the remaining five cannulated peripherally (PC). The median age was 104.5 months (IQR: 17–166.75); 55 (IQR: 17–133) for CC; and 151 months (IQR: 123–193) for PC ($p=0.34$). One of the eight CC patients had a pre-existing medical condition prior to ECMO, whereas all of the PC patients had significant co-morbidities (see Table 1).

Pre-ECMO clinical parameters are given in Table 2. There was no difference between CC and PC patients in pre-ECMO VIS, highest lactate, or INR. The median pre-ECMO white blood cell count was lower in CC than PC patients (9.4 vs $14.9 \times 10^9/L$, $p<0.05$). The use of continuous renal replacement therapy was similar in the two groups, but more patients received plasmapheresis in the CC group.

The maximum flow achieved during VA ECMO (in mL/kg/min) was not significantly different between CC and PC patients, and no difference was observed between the survivors and non-survivors either (Table 3). However, when this maximum flow is corrected to a percentage according to calculations using individual patient's BSA and age, the median maximum flow% achieved in the CC group (179.6%, IQR: 154.4–188.8) was higher than the flow in the PC group (133.5%, IQR: 98.1–149.1), $p=0.01$. This difference was not observed between the survivors and non-survivors. In terms of ECMO duration, the median for the total cohort was 147.1 hours (IQR: 91.9–178.6); the CC and PC groups did not differ in their median ECMO length. There was also no difference in median ECMO duration, ICU, or hospital LOS between survivors and non-survivors.

The majority of patients (10/14) experienced bleeding complications during ECMO (Table 1). In terms of transfusion requirements, the total cohort received a mean of 14.5 (8.5) mL/kg/ECMO day of red blood cells (RBC), 14.5 (9.9) mL/kg/ECMO day of platelets, 5.8 (6.7) mL/kg/ECMO day of fresh frozen plasma (FFP), and 0.3 (1.0) mL/kg/ECMO day of cryoprecipitate (Table 4). There was no difference between the CC and PC groups in their RBC and FFP requirements, but the PC patients received lower median volumes of platelets than the CC patients (3.3 vs 15.6 mL/kg/ECMO day, $p<0.05$).

All patients in the central cohort had the minimum possible FSS of six prior to ECMO, whereas the peripheral patients had higher baseline FSS. Survivors uniformly had a mean increase in their FSS at discharge, as well as the need for rehabilitation services, with 4/8 requiring inpatient rehabilitation. Further details and CNS imaging changes are reported in Table 5.

Survival to ECMO decannulation was 77.8% (7/9) in the central cohort, with a survival to hospital discharge of 66.7% (6/8). In the peripheral group, there was a 40% (2/5) overall survival to decannulation and discharge. Four of the CC patients (44.4%) received mediastinal re-exploration while on ECMO. However, no patient subsequently developed mediastinitis.

Discussion

Our review supports the adoption of a paradigm for ECMO in septic shock that includes central cannulation for selected patients, if appropriate institutional resources exist. In our series, CC was deployed safely with survival to hospital discharge in 66.7% compared to 40% in PC patients, and without significantly increased complications. We would suggest that any comparison in the survival of the two groups be made with caution as the PC group typically had additional

Table 1. Patient demographics.

Patient	Age at ECMO	Gender	Etiology	Pre-existing conditions	Notes	Cannula size (French)	Decannulated from ECMO	Alive at discharge	Cause of death	Bleeding complications	Adjunctive therapies		
											CRRT	TPE	
Central cannulation													
1	11 years old	F	Cervical lymphadenitis	None		18 aorta, 28 RA	Y	Y		Cannula site oozing	N	N	
2	14 years old	M	Culture negative septic shock, hyperosmolar hyperglycemic nonketotic coma	None		22 aorta, 36 RA	Y	Y		None	Y	Y	
3	17 months	M	Streptococcus pneumonia	None	Central VA→VV	14 aorta, 22 RA	Y	Y		Mediastinal bleed, subdural hematomas	Y	Y	
4	12 months	F	MSSA toxic shock syndrome	None	E-CPR→central VA	16 aorta, 24 RA	Y	N	Severe HIE, withdrawn	Pulmonary hemorrhage	N	Y	
5	4 years old	F	Atypical Kawasaki	None	Peripheral→central VA	16 aorta, 26 RA, 16 LA vent	Y	Y		Abdominal hematoma	N	Y	
6	18 months	F	Streptococcus pneumonia bacteremia	None		14 aorta, 22 RA	Y	Y		Cannula site, chest tube bleeding	Y	Y	
7	10 months	F	MSSA neck abscess and mediastinitis	None		14 aorta, 24 RA, 14 LA vent	N	N	Bleeding and necrotic neck and mediastinal tissues at debridement, withdrawn	Necrotic neck tissues, attempted debridement leading to withdrawal	N	Y	
8	15 years old	M	Clostridium septicum	Congenital neutropenia	E-CPR→central VA	22 aorta, 34 RA	N	N	No recovery of cardiac function, withdrawn	None	N	N	
9	7 years old	F	Group A streptococcus	None		18 aorta, 28 RA	Y	Y		Oozing from multiple sites	Y	N	
Peripheral cannulation													
10	16 years old	F	Toxic shock syndrome	Panhypopituitarism s/p craniopharyngioma resection		27 LFV, 19 RFA	Y	Y		Cannula site oozing, epistaxis	N	N	
11	10 years old	F	Toxic shock syndrome	Trisomy 21, PHTN, unbalanced AV canal s/p Fontan	Peripheral VA→Impella	15 LFA, 19 RFV	Y	Y		Cannula site oozing, GI hemorrhage	Y	N	
12	17 years old	F	Culture negative septic shock	Prader Willi, morbid obesity		25 RFV, 19 LFA	N	N	Hemorrhage	Oral and line site oozing, menorrhagia, exsanguination event leading to death on ECMO, bleeding source unclear	Y	N	
13	12 years old	F	Culture negative septic shock	s/p lung transplant		21 RFV, 14 RCCA	N	N	Massive intracranial hemorrhage, withdrawn	Massive intracranial hemorrhage leading to discontinuation of ECMO	N	N	
14	7 months	M	Culture negative septic shock	TGA, DILV s/p PA band		10 RCCA, 14 RIJ	N	N		None	Y	N	

Table 2. Clinical and laboratory data.

Variables of interest	Central cannulation, mean $\pm$ SD, median $\pm$ IQR	Peripheral cannulation	p-Value	Survivors	Non-survivors	p-Value
Pre-ECMO data						
VIS	46.6 $\pm$ 33.2, 45.4 (27–70)	40.1 $\pm$ 16.5, 43.4 (30–46.7)	0.84	41.3 $\pm$ 26.4, 36.7 (24.9–55)	48.1 $\pm$ 13.0, 46.0 (33.8–58.0)	0.70
Highest arterial lactate (mmol/L)	7.6 $\pm$ 4.9, 5 (3.8–13.5)	6.8 $\pm$ 7.8, 13.2 (1.5–11.1)	0.35	7.3 $\pm$ 4.7, 4.5 (3.8–11.7)	7.4 $\pm$ 7.5, 4.8 (2.0–12.4)	0.85
WBC ($\times 10^3/\mu\text{L}$)	8.8 $\pm$ 4.6, 9.4 (6.9–12.9)	7.2 $\pm$ 3.8, 14.9 (12.4–17.0)	0.03	12.0 $\pm$ 3.8, 13.2 (8.8–14.7)	11.4 $\pm$ 10.7, 10 (5.3–11.9)	0.41
Platelet ($\times 10^3/\mu\text{L}$)	73.2 $\pm$ 58.7, 63 (28–101)	127 $\pm$ 128.2, 88 (46–112)	0.42	61.3 $\pm$ 40.5, 69.5 (23.8–91.3)	134 $\pm$ 120.9, 86 (50.2–175)	0.27
D-dimer* ($\mu\text{g/mL FEU}$)	14.1 $\pm$ 7.0, 20 (7.2–20)	9.3 $\pm$ 4.4, 4.3 (1.33–20)	0.23	12.5 $\pm$ 8.2, 14.4 (5.5–0)	12.1 $\pm$ 9.0, 13.6 (5.8–20)	0.95
INR	2.1 $\pm$ 0.6, 1.9 (1.7–2.4)	1.7 $\pm$ 0.2, 1.5 (1.5–1.6)	0.20	2.1 $\pm$ 0.7, 1.9 (1.6–2.6)	1.8 $\pm$ 0.5, 1.6 (1.4–2)	0.37
Avg UOP 8 hours pre-cannulation (mL/kg/hour)	0.8 $\pm$ 0.7, 0.8 (0.2–1.4)	0.9 $\pm$ 0.4, 0.4 (0.3–1.5)	0.64	0.84 $\pm$ 0.73, 0.76 (0.2–1.4)	0.9 $\pm$ 0.9, 0.61 (0.3–1.3)	0.70
Other clinical variables						
ICU length of stay (days)	27 $\pm$ 31.5, 17 (13–24)	34.6 $\pm$ 35.4, 17 (9–50)	0.84	36.9 $\pm$ 31.2, 22.5 (17–45.5)	25.5 $\pm$ 34.0, 8 (6.3–30)	0.17
Hospital length of stay (days)	44.6 $\pm$ 39.0, 38 (32–50)	37.8 $\pm$ 34.8, 29 (9–54)	0.94	63.9 $\pm$ 39.4, 50.5 (41.8–68.5)	25.5 $\pm$ 34.0, 8 (6.3–30)	0.06

*D-dimer values of >20 were imputed to be 20 for purposes of calculation.

Bold font indicates statistical significance.

Table 3. ECMO specific variables.

ECMO variables	Central cannulation, mean $\pm$ SD, median $\pm$ IQR	Peripheral cannulation	p-Value	Survivors	Non-survivors	p-Value
Max ECMO flow through VA run (mL/kg/min)	198 $\pm$ 88, 176 (115–258)	105.9 $\pm$ 57.6, 90.2 (87.6–117.4)	0.06	183.1 $\pm$ 100.9, 145.8 (95.8–268.6)	142.1 $\pm$ 71.4, 139.3 (105.3–186.9)	0.75
% Max flow indexed to CI expected flow*	183.8 $\pm$ 47.2, 179.6 (154.4–188.1)	125.4 $\pm$ 26.2, 133.5 (98.1–149.1)	0.01	179.1 $\pm$ 56, 167.0 (149.8–201.4)	141.5 $\pm$ 30.8, 141.3 (125.9–154.8)	0.22
Time to vasopressor discontinuation after ECMO start (hours)	10.6 $\pm$ 11.5, 6 (0–18)	33 $\pm$ 38.1, 14 (12–41)	0.26	12.5 $\pm$ 14.7, 8.5 (1.5–19.5)	26.3 $\pm$ 35.6, 13 (7.5–26)	0.65
ECMO duration† (hours)	209.1 $\pm$ 174.5, 148.1 (134–196)	105.0 $\pm$ 66.5, 77.8 (67.7–167.5)	0.35	194.6 $\pm$ 174.4, 147.1 (118.5–185.7)	141.6 $\pm$ 121.3, 114.7 (63.7–163.5)	0.56
VIS @ 6 hours	4.6 $\pm$ 6.2, 3.8 (0–5)	5.6 $\pm$ 7.1, 2 (0–10)	1	4.2 $\pm$ 5.3, 3.7 (0–5)	6.0 $\pm$ 7.8, 2.9 (0.5–8.4)	0.90
VIS @ 24 hours	2.4 $\pm$ 3.6, 0 (0–3.8)	8.3 $\pm$ 8.4, 8.5 (1.5–15.3)	0.5	2.8 $\pm$ 5.5, 0 (0–2.8)	5.4 $\pm$ 6.1, 3.7 (0.5–9.1)	0.40
Vasodilators use (excluding milrinone) within 12 hours of cannulation	66.7% (6/9)	40% (2/5)	0.33	62.5% (5/8)	33.3% (2/6)	0.28

*Expected cardiac index calculated as a constant based on age $\times$ BSA.

†Only VA run included in calculations.

Bold font indicates statistical significance.

Table 4. Blood product requirements.

Blood product (mL/kg/ECMO day)	Central cannulation, mean $\pm$ SD, median (IQR)	Peripheral cannulation	p-Value	Survivors	Non-survivors	p-Value
RBC	16.4 $\pm$ 7.3, 13.7 (9.7–24.1)	11.0 $\pm$ 4.6, 8.9 (4.1–12.8)	0.14	15.3 $\pm$ 6.9, 13.6 (9.6–17.6)	13.4 $\pm$ 7.2, 10.9 (5.3–9.6)	0.48
Platelet	18.7 $\pm$ 8.9, 15.6 (14.7–22.5)	7.0 $\pm$ 3.2, 3.3 (2.4–8.7)	0.03	16.4 $\pm$ 6.1, 15.5 (14.0–21.1)	12.0 $\pm$ 13.8, 6.0 (2.6–16.1)	0.22
FFP	4.7 $\pm$ 3.5, 3.3 (2.8–8.2)	7.7 $\pm$ 4.8, 1.5 (1.6–10.3)	0.4	4.9 $\pm$ 3.7, 3.3 (2.7.4/8)	7.2 $\pm$ 9.8, 3.1 (1.3–8.8)	0.75

Bold font indicates statistical significance.

comorbidities that would have (in our institution) precluded a central approach

While ECMO *per se* now has an established place in the management of refractory septic shock, the role of central VA ECMO remains controversial. This likely relates to logistic factors that vary between institutions, as well as the perceived increased risk of the central approach. Indeed in our center, peripheral cannulation is almost always performed by pediatric surgeons in the PICU, and central cannulation—by cardiac surgeons, in the cardiac operating rooms. Furthermore, while all patients with septic shock would typically be in the PICU prior to ECMO, those who have been centrally cannulated are moved to the CICU once on ECMO to have more immediate availability of the cardiac surgical team, and also familiarity with managing immediate post-sternotomy (and open sternum) by medical and nursing teams. This requires excellent collaboration and coordination between medical and surgical teams. The perceived risks refer primarily to the potential for increased bleeding and need for re-exploration in patients who are already coagulopathic, as well as the perception that patients will develop mediastinitis as a result. In our experience which closely resembles that of Melbourne,¹⁴ the need for re-exploration for CC patients was not uncommon, but mediastinitis did not complicate their recovery. In terms of blood product requirements—similar to Melbourne—there was no difference in RBC and FFP transfusions, though our CC patients did require higher daily volumes of platelets. The reason for this could in part relate to institutional thresholds for transfusions (ours typically being 100 for this group), and potentially a more liberal approach given that our center was on an early part of its learning curve.

While there are promising reports, a major limitation in employing VA ECMO in cases of vasoplegic shock appears to be achieving a flow that can meet the patients' metabolic demand. The dearth of high quality evidence available also makes committing to this particular therapy a daunting proposition.

We would suggest that our case series provides further experience to support the proposed role of central VA ECMO for refractory septic shock. As a comparison to usual practice, a query of the ELSO database for pediatric septic shock in the same time period revealed a total of 202 patients placed on VA ECMO, with 69 of those patients going onto ECMO after E-CPR. Survival

to hospital discharge in the overall population was 28.2% (57/202), with much lower numbers as expected for the E-CPR recipients (14.4%, 10/69). Out of this ELSO cohort, 34 patients were identified as having been cannulated centrally with 13 (38%) patients surviving to discharge. While our cohort is small, our center managed to achieve a 77.8% survival to ECMO decannulation in the CC patients, with a 66.7% survival to discharge, which suggests better outcomes are possible. The mortality in our centrally cannulated patients were through withdrawal of life sustaining therapy further in their course of illness, implying that we are able to adequately sustain patients through their initial stages of shock using a central cannulation strategy.

Consistent with other reports,^{13,14} the time between presentation to the PICU and ECMO cannulation was relatively short with a median of 15 hours. This requires early recognition of refractory shock as well as a proactive approach with early consideration of ECMO, and the ability to deploy this quickly. Multiple studies have reported that up to more than 50% of the mortality for children referred to the PICU for sepsis happen in the first 24 hours. Thus, the ability to support patients early with central cannulation may have contributed to our success rate as we are able to provide adequate hemodynamic support prior to the establishment of more severe and potentially irreversible organ dysfunction.

While “mL/kg/min” was utilized in the previous pediatric ECMO studies, this may not be the best approach to describe the physiologic impact of ECMO flow within the entire pediatric population that widely varies in terms of body size and thereby metabolic demand. Many pediatric heart centers, including ours, have their own flow scale for cardiopulmonary bypass that is not static (e.g. ## mL/kg/min) across ages rather varies depending on the patient age/size.¹⁶ This is why it may be more appropriate to express the ECMO flow according to a percentage to the standard full flow (which is defined as a minimum flow requirement to meet metabolic demand that is maximally reduced with general anesthesia/deep sedation). Not surprisingly, the ECMO flow is significantly higher with the central approach than in the peripheral group. The difference in ECMO flow would likely have impacted hemodynamics early after ECMO initiation, which would include time to vasoconstrictor discontinuation (8.9 vs 13 hours) and vasodilator use within 12 hours of ECMO support (67%

Table 5. Outcomes FSS.

Patient	FSS on admission	FSS at discharge	CNS imaging findings	Other complications	ICU LOS (days)	Hospital LOS (days)	Rehab at discharge
Central							
1	6	8	MRI: mild volume loss		17	45	Yes—inpatient
2	6	7	MRI: minor volume loss, possible central pontine myelinolysis, questionable small hemorrhage	Left toes with necrosis, left foot drop	15	51	Yes—inpatient
3	6	13	MRI: small bilateral subdural hemorrhage, volume loss	Necrosis to multiple fingers	107	138	Yes (PT/OT outpatient)
4	6	Deceased	MRI: severe HIE		37	37	N/A
5	6	9	MRI: mild volume loss		21	32	Yes (PT/ST/OT outpatient)
6	6	10	N/A	Bilateral toes with necrosis	24	50	Yes (ST/OT outpatient)
7	6	Deceased	CT: small acute subarachnoid hemorrhage along right parietal sulci		6	6	N/A
8	6	Deceased	CT: hypodensity from right hippocampus, potentially from hypoxic event		4	4	N/A
9	6	10	N/A	Right BTK and left foot amputation	13	38	Yes—inpatient
Peripheral							
10	11	11	MRI: prominent lateral ventricles, unchanged from previous		17	29	Yes—inpatient
11	10	12	CT: interval volume loss, prominent ventricles, sulci, and cisterns		50	54	Yes (PT/OT outpatient)
12	8	Deceased	CT: severe cerebral edema, uncal herniation, possible HIE		7	7	N/A
13	8	Deceased	CT: massive intraparenchymal hemorrhage centered in the left frontal lobe, with extension into the ventricular system. Generalized mass effect including subfalcine and tonsillar herniation		9	9	N/A
14	6 (newborn)	Deceased	N/A		90	90	N/A

vs 20%). Although these clinically significant differences did not reach a level of statistical significance, such impacts may not be ignored, given the time-sensitive nature of this grave condition (>50% mortality within 24 hours of PICU admission^{3,21}).

Finally, our center also employs other modes of extracorporeal therapy such as CRRT^{22,23} and TPE^{24,25} liberally, which may have contributed to good outcomes.

Although survival to discharge was high, it is notable that all of the survivors had increased FSS and required rehabilitation. One patient was discharged with a tracheostomy and was ventilator dependent; the patients who received CRRT had varying degrees of renal recovery with only one out of three going on to require a renal transplant. While the discharge morbidity in our cohort

is significant, this is consistent with the trend of increasing morbidity amongst PICU survivors, especially those admitted with high severity of illness.^{26,27} MRI of the brain was available on all patients and the four survivors had at least minimal volume loss. Outcomes of particular interest were intracranial hemorrhage and ischemia, given the inherent hemorrhagic risks of ECMO. However, only one patient demonstrated small subdural hemorrhages in his CNS imaging, despite the derangements in the clotting milieu that accompany cases of advanced septic shock. Of note, these outcomes may be skewed by the fact that they are reported at the time of hospital discharge and not at a later time point.

There are several limitations to this study; most notably, its retrospective nature and small cohort, and prior

to the start time of our report, VA ECMO was only infrequently used to support sepsis in our institution. Additionally, the caveat stated earlier that our two “subgroups” are not equivalent, so outcomes comparisons, with the exception of ECMO flow and its physiologic impact, complications, or transfusion requirement, should be made with caution. Ours is a descriptive report with the goal of describing the results of our central cannulation program and providing additional data to inform other institutions, or potentially future studies. The program is a result of an extensive multidisciplinary and collaborative effort amongst intensivists (pediatric and cardiovascular), surgeons (pediatric general/thoracic and congenital heart), perfusionists, and ECMO specialists. Over the years, the algorithm has been iteratively refined and streamlined, allowing for a more efficient activation and implementation of central cannulation for these critically-ill children.

ECMO candidacy selection has been and likely will remain a subject that generates much discussion, but our data suggests that central VA ECMO in refractory septic shock is a modality that can have good outcomes. While the central cannulation approach is labor intensive at the outset, this technique should be strongly considered for previously healthy children who had otherwise failed all other conventional therapy.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Amanda Ruth  <https://orcid.org/0000-0003-4914-7559>

Supplemental material

Supplemental material for this article is available online.

References

1. Weiss SL, Peters MJ, Alhazzani W, et al. Surviving sepsis campaign international guidelines for the management of septic shock and sepsis-associated organ dysfunction in children. *Pediatr Crit Care Med* 2020; 21: e52–e106.
2. Boeddha NP, Schlapbach LJ, Driessen GJ, et al. Mortality and morbidity in community-acquired sepsis in European pediatric intensive care units: a prospective cohort study from the European Childhood Life-threatening Infectious Disease Study (EUCLIDS). *Crit Care* 2018; 22: 143.
3. Weiss SL, Balamuth F, Hensley J, et al. The epidemiology of hospital death following pediatric severe sepsis: when, why, and how children with sepsis die. *Pediatr Crit Care Med* 2017; 18: 823–830.
4. Kutko MC, Calarco MP, Flaherty MB, et al. Mortality rates in pediatric septic shock with and without multiple organ system failure. *Pediatr Crit Care Med* 2003; 4: 333–337.
5. Lin JC, Spinella PC, Fitzgerald JC, et al. New or progressive multiple organ dysfunction syndrome in pediatric severe sepsis: a sepsis phenotype with higher morbidity and mortality. *Pediatr Crit Care Med* 2017; 18: 8–16.
6. Rhodes A, Evans LE, Alhazzani W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock: 2016. *Intensive Care Med* 2017; 43: 304–377.
7. Brierley J, Carcillo JA, Choong K, et al. Clinical practice parameters for hemodynamic support of pediatric and neonatal septic shock: 2007 update from the American College of Critical Care Medicine. *Crit Care Med* 2009; 37: 666–688.
8. Davis AL, Carcillo JA, Aneja RK, et al. American College of Critical Care Medicine clinical practice parameters for hemodynamic support of pediatric and neonatal septic shock. *Crit Care Med* 2017; 45: 1061–1093.
9. Weiss SL, Peters MJ, Alhazzani W, et al. Surviving sepsis campaign international guidelines for the management of septic shock and sepsis-associated organ dysfunction in children. *Intensive Care Med* 2020; 46: 10–67.
10. McCune S, Short BL, Miller MK, et al. Extracorporeal membrane oxygenation therapy in neonates with septic shock. *J Pediatr Surg* 1990; 25: 479–482.
11. Goldman AP, Kerr SJ, Butt W, et al. Extracorporeal support for intractable cardiorespiratory failure due to meningococcal disease. *Lancet Lond Engl* 1997; 349: 466–469.
12. Oberender F, Ganeshalingham A, Fortenberry JD, et al. Venoarterial extracorporeal membrane oxygenation versus conventional therapy in severe pediatric septic shock. *Pediatr Crit Care Med* 2018; 19: 965–972.
13. Maclaren G, Butt W, Best D, et al. Extracorporeal membrane oxygenation for refractory septic shock in children: one institution's experience. *Pediatr Crit Care Med* 2007; 8: 447–451.
14. MacLaren G, Butt W, Best D, et al. Central extracorporeal membrane oxygenation for refractory pediatric septic shock. *Pediatr Crit Care Med* 2011; 12: 133–136.
15. Schlapbach LJ, Chiletto R, Straney L, et al. Defining benefit threshold for extracorporeal membrane oxygenation in children with sepsis—a binational multicenter cohort study. *Crit Care* 2019; 23: 429.
16. Adachi I, Shekerdemian L, Checchia PA, et al. Chapter 75. Treatment of heart failure: mechanical support. In: *Roger's textbook of pediatric intensive care*. 5th ed. Wolters Kluwer, 2015.
17. Pollack MM, Holubkov R, Glass P, et al. Functional status scale: new pediatric outcome measure. *Pediatrics* 2009; 124: e18–e28.
18. Pollack MM, Holubkov R, Funai T, et al. Relationship between the functional status scale and the pediatric overall performance category and pediatric cerebral performance category scales. *JAMA Pediatr* 2014; 168: 671–676.

19. Cashen K, Reeder R, Dalton HJ, et al. Functional status of neonatal and pediatric patients after extracorporeal membrane oxygenation. *Pediatr Crit Care Med* 2017; 18: 561–570.
20. Pereira GA, Schaan CW, Ferrari RS. Functional evaluation of pediatric patients after discharge from the intensive care unit using the Functional Status Scale. *Rev Bras Ter Intensiva* 2017; 29: 460–465.
21. Cvetkovic M, Lutman D, Ramnarayan P, et al. Timing of death in children referred for intensive care with severe sepsis: implications for interventional studies. *Pediatr Crit Care Med* 2015; 16: 410–417.
22. Selewski DT, Cornell TT, Blatt NB, et al. Fluid overload and fluid removal in pediatric patients on extracorporeal membrane oxygenation requiring continuous renal replacement therapy. *Crit Care Med* 2012; 40: 2694–2699.
23. Gorga SM, Sahay RD, Askenazi DJ, et al. Fluid overload and fluid removal in pediatric patients on extracorporeal membrane oxygenation requiring continuous renal replacement therapy: a multicenter retrospective cohort study. *Pediatr Nephrol* 202017; 35: 871–882.
24. Fortenberry JD, Nguyen T, Grunwell JR, et al. Therapeutic plasma exchange in children with thrombocytopenia-associated multiple organ failure: the thrombocytopenia-associated multiple organ failure network prospective experience. *Crit Care Med* 2019; 47: e173–e181.
25. Kawai Y, Cornell TT, Cooley EG, et al. Therapeutic plasma exchange may improve hemodynamics and organ failure among children with sepsis-induced multiple organ dysfunction syndrome receiving extracorporeal life support. *Pediatr Crit Care Med* 2015; 16: 366–374.
26. Farris RWD, Weiss NS, Zimmerman JJ. Functional outcomes in pediatric severe sepsis: further analysis of the researching severe sepsis and organ dysfunction in children: a global perspective trial. *Pediatr Crit Care Med* 2013; 14: 835–842.
27. Pinto NP, Rhinesmith EW, Kim TY, et al. Long-term function after pediatric critical illness: results from the survivor outcomes study. *Pediatr Crit Care Med* 2017; 18: e122–e130.